

STATEMENT OF WORK
Mechanism Enhancement Reconditioning and Refurbishment of Eaton/Cutler Hammer
150VCP-WG63 Circuit Breaker for Detroit

- 1.0 General Statement** The Contractor shall provide all personnel, equipment, tools, materials, supervision, and other items and services necessary to install the mechanism enhancements, and repair damage to the flex leads on the circuit breaker as defined in this Statement of Work (SOW). The circuit breaker is an Eaton / Cutler Hammer brand, 150VCP-WG63 circuit breaker with serial number 02060877.

The Contractor shall perform to the standards in the contract and comply with all local, state, and federal regulations.

This is a firm fixed price contract.

2.0 Definitions

Contracting Officer (CO): The Government employee who is authorized to enter into, administer, and/or terminate contracts and make related determinations and findings.

American National Standards Institute (ANSI)

Institute of Electrical and Electronics Engineers (IEEE)

Willamette Valley Project (WVP): The destination of where the contract is to take place.

3.0 Applicable Documents:

ANSI/IEEE C37.04 – Standard Rating Structure for AC HV Circuit Breakers

ANSI/IEEE C37.06 – Preferred Ratings for AC HV Circuit Breakers

ANSI/IEEE C37.09 – Standard Test Procedure for AC HV Circuit Breakers

I.B. 32-255-1G – Eaton "Instructions for Installation, Operation and Maintenance of Type VCP-W Vacuum Circuit Breakers".

4.0 Government Will Furnish:

One (1) Eaton / Cutler Hammer 150VCP-WG63 3000A Circuit Breaker in need of refurbishment.

5.0 Contractor Shall Furnish:

All manpower, material, equipment, and supervision necessary to perform the requirements of this contract.

6.0 Background:

The Eaton / Cutler Hammer type 150VCP-WG63 vacuum circuit breaker serves as a main unit breaker between the main generator and step up transformers at the Detroit Dam, in Detroit OR.

The copper flex leads on all three phases of the circuit breaker have sustained damage from breaker operating mechanism over travel.

The circuit breaker has not received any of the Factory Authorized Mechanism Enhancements such as the damper for the operating mechanism which prevents this over travel issue.

7.0 Nameplate Information:

All parts and repairs must be compatible with the Eaton type 150VCP-WG63 circuit breaker. Parts shall be rated to meet or exceed the performance ratings specified on the breaker nameplate, given as:

Manufacturer:

Eaton / Cutler Hammer

Type:

150VCP-WG63

Maximum Voltage (V):	15,000V
Voltage Range Factor (K):	1.0
Power Frequency:	60 Hz
Continuous Current:	3000 A
Lightning Imp. Withstand Voltage:	95 kV peak
Interrupting Time:	83 ms
Short-Circuit Current (I):	63kA rms
Serial Number:	02060877
Weight:	525 lbs.
Instruction Book Number:	32-255-1F
Parts List (Style) Number:	4A35145C26
Option Number:	20210200010000000000
Wiring Diagram Number:	691C386H44

8.0 Scope:

This SOW covers replacement of vacuum interrupters, replacement or repair of all three copper flex leads, installation of the factory authorized mechanism enhancements, and calibration and testing of one (1) Eaton / Cutler Hammer type 150VCP-WG63 vacuum circuit breaker. Satisfactory completion of this work requires that parts be furnished and installed per factory procedures to restore the breakers to original factory performance standards as verified by testing performed in accordance with the factory's operation and maintenance manual (I.B. 32-255-1G – Eaton "Instructions for Installation, Operation and Maintenance of Type VCP-W Vacuum Circuit Breakers").

9.0 Requirements:

9.1 Repair and Reconditioning:

- 9.1.1** Replace all three vacuum pole assemblies with new assemblies of brand name or equal to superseding type meeting IEEE/ANSI C37.04 and C37.06 standards. Existing vacuum pole assemblies are Cutler Hammer Brand, Type WL35123C.
- 9.1.2** Replace or repair of all three damaged flex leads between vacuum interrupter assembly and breaker stabs.
- 9.1.3** Installation of Factory Authorized Mechanism Enhancements including the operating mechanism damper. Clarification: Factory Authorized Mechanism Enhancements (FAME) is a reference to the objective of the installation of a damper device / dashpot assembly which will prevent the moving contact arm from impacting the copper flex leads. If the proposal includes the installation of the spring damper mechanism and is then installed and tested to factory / manufacturer standards as laid out in SOW 9.2, then the proposal is acceptable.
- 9.1.4** Factory calibration and of circuit breaker following vacuum pole assembly replacement.

9.2 Testing:

Upon completion of all required work, circuit breaker shall be tested and perform to manufacturer's specifications. Testing shall follow ANSI/IEEE C37.09 test procedures and include, at a minimum, insulation resistance, contact resistance, timing, and over-potential tests. A report with the test results and certification that the breaker meets Eaton's performance specifications and that it is suitable for safe operation shall be submitted to the Government.

10.0 Warranty

A 12 month warranty will be provided to the Government for the 150VCP-WG63 circuit breaker.

11.0 Shipping

The circuit breaker is currently in the possession of U.S. Army Corps of Engineers, Willamette Valley Project, and located at the Detroit Dam, in Detroit OR. The Contractor shall arrange for transport of the circuit breaker from its current location to the contractor's facility. After repairs are complete and satisfactory test results are submitted to the Government, the breaker shall be delivered free on board (F.O.B.) to Detroit Dam. The Government shall not be responsible for damage or loss of the circuit breaker during shipping or while being stored away from Detroit Dam. The Government shall not be responsible for the cost of shipping to the contractor facility or shipping back to Detroit Dam. After repairs are completed, delivery to Detroit Dam is to be scheduled during the hours of 7am to 4pm Monday through Thursday only.

Detroit Dam Address:

U.S. Army Corps of Engineers
Detroit Dam
40990 Hwy 22
Mill City, OR 97360

12.0 Government Acceptance.

12.1 Acceptance Requirements Summary. These are the criteria against which the Government acceptance will be determined.

CONTRACT REQUIREMENTS SUMMARY				
Objective	Contract Requirement	Allowed Number of Discrepancies	Surveillance	Validated Complaints
Shipping without damage	Breaker received clean without shipping damage	0	visual inspection	
Replacement of pole units	New vacuum poles installed having new serial numbers	0	Confirm new serial numbers of replacement bottles	
Replacement or repair of dented flex leads	All three flex leads between vacuum interrupter assemblies and breaker stabs are repaired or replaced	0	Confirm no dents on flex leads	
Installation of Factory Authorized Mechanism Enhancements	The operating mechanism damper has been installed	0	Confirm installation of damper	
Spring charging mechanism operates properly and reliably	No malfunction of spring charging is observed during commissioning.	0	Test installation	
Open and Close Operation is proper and reliable	No malfunction of open or close operation is observed during commissioning.	0	Test installation	

13.0 GENERAL

13.1 Points of Contact.

Contracting Officer:
Email:
Office Phone:

Contract Specialist:
Email:
Office Phone: